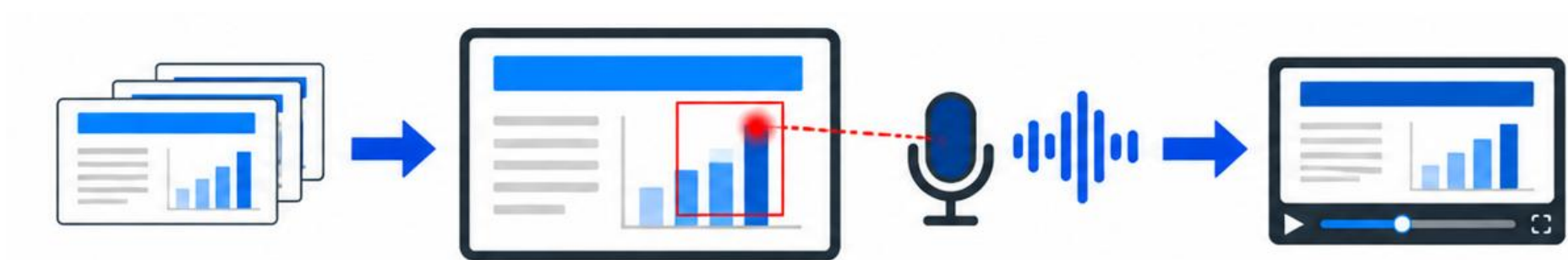


A Slides-to-Video Skill with Guided Visual Attention

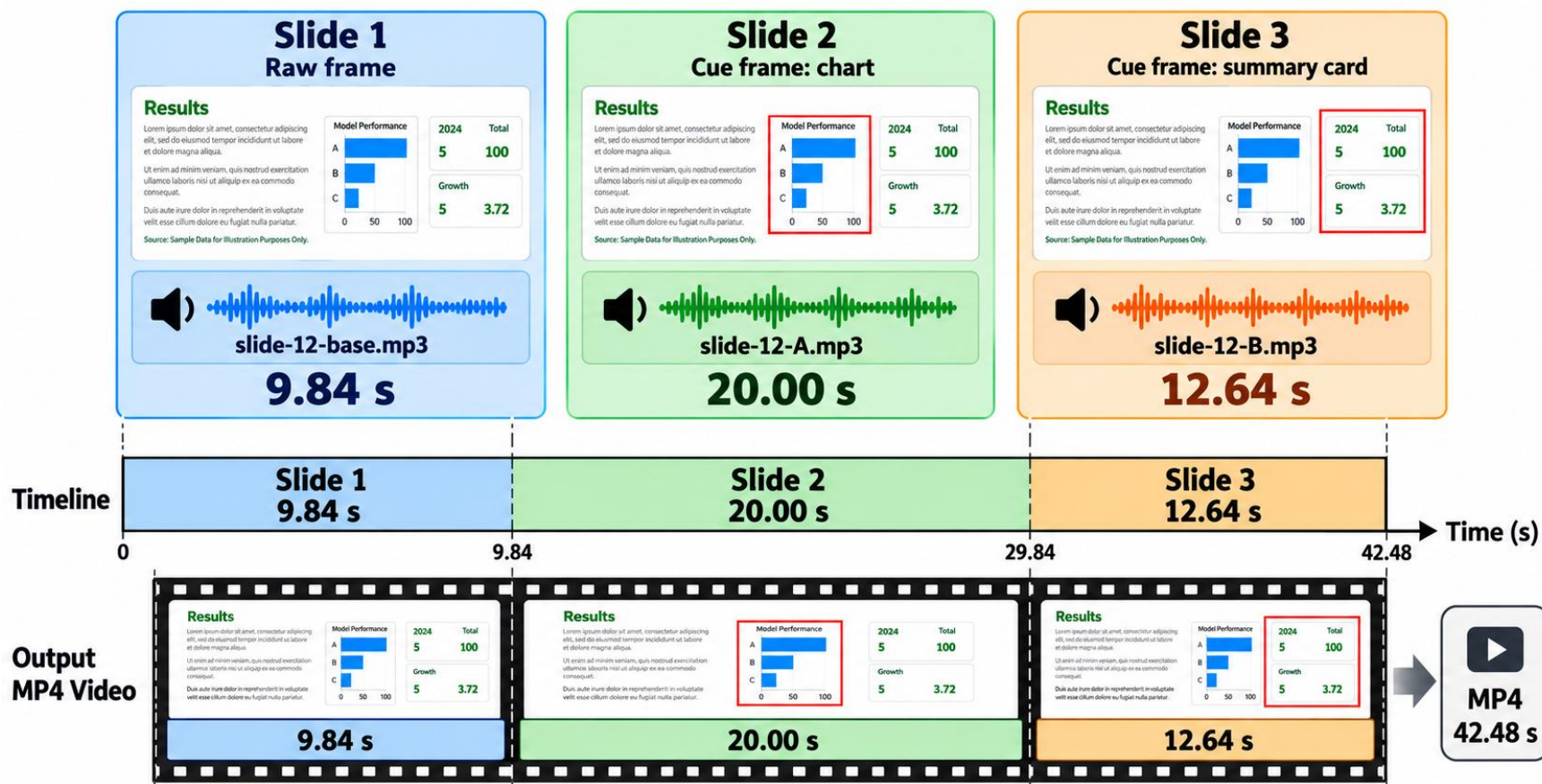
带视线引导的幻灯片视频生成 Skill

The narration explains the slide while visual cues show the viewer where to look
旁白负责讲解内容，视觉提示同步引导观众看向对应区域



把一份静态幻灯片变成带有视觉提示和旁白的视频 Transform a static slide deck into a video featuring visual cues and narration.

- 1 讲稿写什么**
写什么内容、按什么顺序讲
- 2 让观众看哪里 (视觉提示)**
讲到哪里, 画面就要指到哪里
- 3 图与声对齐**
画面停留多久, 服从音频实际长度



演示视频Demo Video

全球AIGC市场规模与增长趋势 (2021-2025)

Global AIGC Market Size and Growth Trends (2021–2025)

图 2-1 2021-2025年全球AIGC历史市场规模 (亿美元)



爆发式增长：技术红利全面释放 (2021-2023)

Explosive Growth: Full Unleashing of the Technology Dividend (2021–2023)



规模化落地：从技术供给到场景驱动 (2024-2025)

Large-Scale Implementation: From Technology Supply to Scenario-Driven Adoption (2024–2025)



中国市场：增速领跑全球，潜力持续释放
Chinese Market: Leading Global Growth, Potential Continues to Be Unleashed

1. 把幻灯片转换成图片

Convert slides into images

1. 把幻灯片转换成图片 Convert slides into images

Why images?

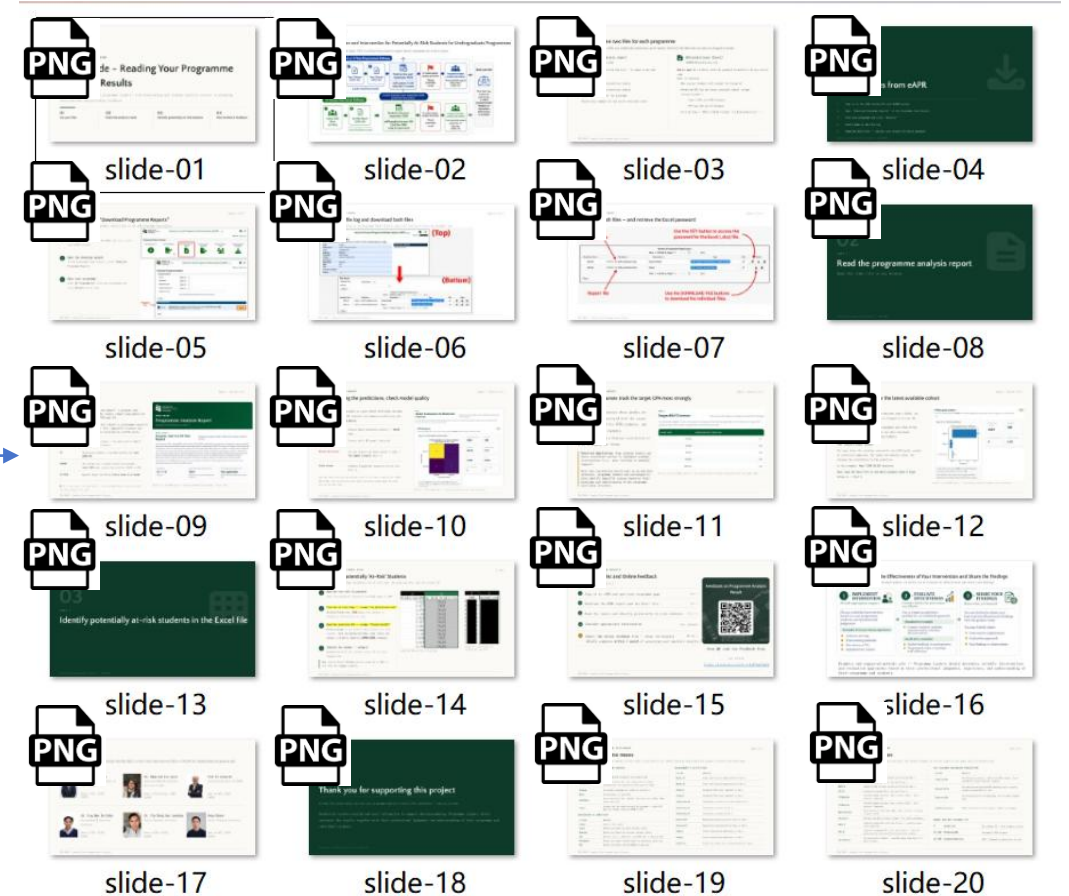
时间轴要的是帧
Timeline needs frames



Slides PPTX (20 Pages)

PPTX->PDF->PNG

Suggested tool:
LibreOffice; pdftoppm



一页幻灯片 = 一张图

One page, one PNG. Fixed resolution

2. 让视觉大模型「看图」，产出视觉锚点

Ask a vision model to mark visual anchors

2. 让视觉大模型「看图」，产出视觉锚点

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PART 2 • READ THE REPORT

Report • Section 4 of 4

Model predictions for the latest available cohort

PREDICTION SUMMARY CARDS

In this example, the cards give the analysed cohort (2024), the number of students (109), how many are flagged as at-risk (5), and the optimal cutoff used (2.72).

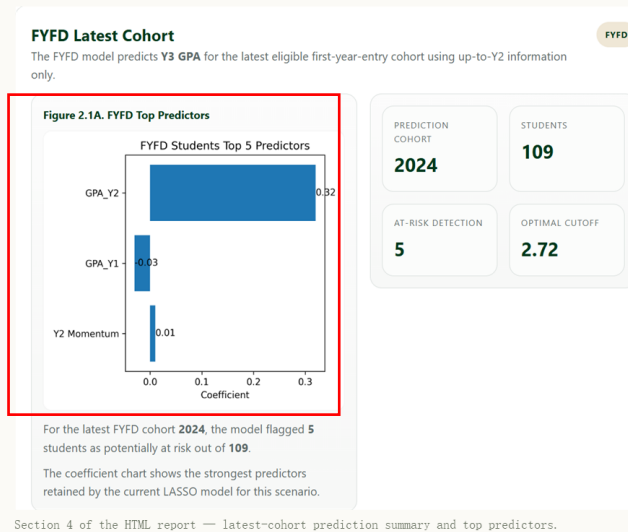
*Note: Predictions provide advisory guidance only. Even if the at-risk detection is 0, PL can still use their firsthand knowledge to spot potentially at-risk students.

TOP PREDICTORS CHART

The chart shows the variables retained by the LASSO model, ranked by coefficient magnitude. The larger the absolute value, the stronger the contribution to the prediction.

In this example, Year 2 GPA (0.32) dominates.

Next: open the Excel file to see which students those 5 flags belong to → Part 3.



- slide: 12

title: "Model predictions for the latest available cohort"

visuals:

- id: A

location: "Right report screenshot, Figure 2.1A"

element: "FYFD Students Top 5 Predictors bar chart —

GPA_Y2 (longest bar, 0.32, dominates), GPA_Y1 0.03, Y2 Momentum 0.01"

shape: rectangle

box: [0.534, 0.435, 0.213, 0.255]

- id: 视觉锚点编号 A / B / C
- location: 在画面上的位置 Where on the slide
- element: 这个区域的内容是什么 what the region is
- shape: 视觉提示的形状 rectangle/dot
- box: 视觉提示的位置坐标信息 The coordinates of the visual cues

Input: Image(PNG)
输入: 每页图片



Output: Visual Anchor Annotation(.yaml)
输出: 视觉锚点标注文件

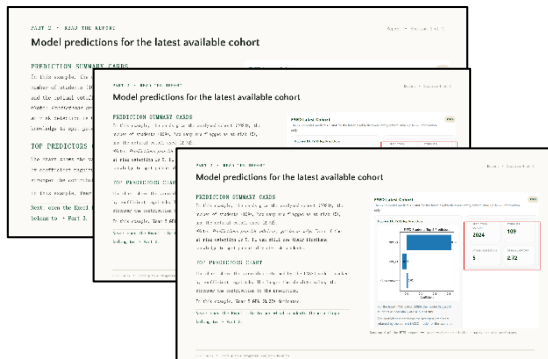
视觉大模型 Vision LLM
(e.g., MIMO; DS-V4.1)

3. 生成带视觉锚点标记的讲稿

Write a script tagged to those anchors

3. 生成讲稿: 根据视觉锚点组织讲解顺序

Build the narration around the visual anchors



Input 1: Images (PNG)
输入1: PNG 图片

- slide: 12
title: "Model predictions for the latest available cohort"
visuals:
- id: A
location: "Right report screenshot, Figure 2.1A"
element: "FYFD Students Top 5 Predictors bar chart — GPA_Y2
(longest bar, 0.32, dominates), GPA_Y1 0.03, Y2 Momentum 0.01"
shape: rectangle
box: [0.534, 0.435, 0.213, 0.255]

Input 2: Visual Anchor Annotation(.txt,,.yaml)
输入2: 视觉锚点标注文件

Script Guide Prompt (.txt, .md)

Narration Format
Techniques about how to write good script
Tone
Structure
Pace
Style

Input 3: Script Guide (.txt,.md)
输入3: 讲稿写作规范

Input 1: Images (PNG)

Input 2: Visual
Anchor Annotation

Input 3: Script Guide

Xiaomi
MIMO



Vision LLM
(e.g., MIMO; DS-V4.1)

Slide 12 — Report • Section 4 of 4 — Predictions for the latest cohort

Role: proof model's accuracy [~38s]

Visuals:

- [A] Top predictors bar chart (LASSO)
- [B] Prediction summary cards

Say: The final report section moves from historical validation to predictions for the latest eligible cohort. [A] The chart explains which variables the LASSO model retained. LASSO is a regression method that shrinks less useful coefficients toward zero, leaving a relatively compact set of predictors. It was chosen in part because the retained variables and their coefficients are easier to explain and audit than a more opaque model such as a neural network. The larger the absolute coefficient, the stronger the retained contribution; here, Year two GPA is clearly the dominant predictor at zero point three two. [B] In this example, the cohort is twenty twenty-four, with one hundred and nine students, and five are flagged as potentially at risk using the model's optimal cutoff of two point seven two

讲稿格式 Script
Format

Page context first,
then cue by cue
先讲整页，再按视觉
锚点锚点展开

4. 根据视觉锚点的坐标画出提示，把一页拆成多帧

Draw cues from coordinates and split one slide into focused frames

PART 2 • READ THE REPORT Report • Section 4 of 4

Model predictions for the latest available cohort

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FYFD Latest Cohort
The FYFD model predicts **Y2 GPA** for the latest eligible first-year cohort using up-to-Y2 information only.

Figure 3.1A: FYFD Top Predictors

PREDICTION COHORT	STUDENTS
2024	109

AT-RISK DETECTION	OPTIMAL CUTOFF
5	2.72

Section 4 of the FYFD report → latest student prediction summary and top predictors.

CSAC 61467 • Tracking Your Programme Academic Results

PART 2 • READ THE REPORT Report • Section 4 of 4

Model predictions for the latest available cohort

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CSAC 61467 • Tracking Your Programme Academic Results

4. 按坐标画出视觉提示，并把一页拆成多帧 Draw cues from coordinates and split one slide into focused frames

- slide: 12

title: "Model predictions for the latest available cohort"

visuals:

- id: A

location: "Right report screenshot, Figure 2.1A"

element: "FYFD Students Top 5 Predictors bar chart — GPA_Y2 (longest bar, 0.32, dominates), GPA_Y1 0.03, Y2 Momentum 0.01"

shape: rectangle

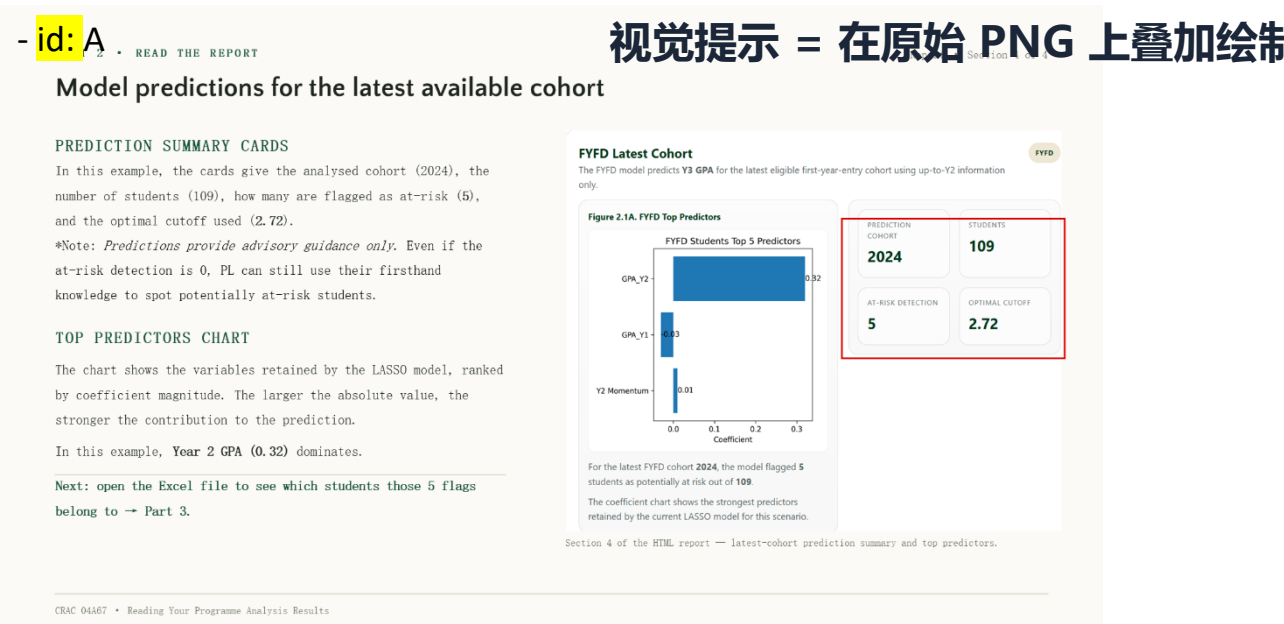
box: [0.534, 0.435, 0.213, 0.255]

怎么画 / How

对于红框提示，用 Python（Pillow）读取原图，按 box [x, y, w, h] 换算成像素。严格使用标注坐标，不重新估计位置。Draw with code. Scale normalized box values by image width and height, then stroke a rectangle. Do not ask a model to paint the cue.

为什么视觉提示由程序画，而不是大模型画 Why the program draws the cue, not the LLM

- Cost and Speed 成本与速度 – 大模型整页重绘更贵更慢
- Unstable Location of Visual Cue 位置不稳定 - 同一坐标每次生成都可能漂移，无法复现
- Risk of Redrawing and Content Change 有重绘风险 - 生成式重绘会改动页面内容与文字



4. 按坐标画出视觉提示，并把一页拆成多帧

Draw cues from coordinates and split one slide into focused frames

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element: "FYFD Students Top 5 Predictors bar chart — GPA_Y2 (longest bar, 0.32, dominates), GPA_Y1 0.03, Y2 Momentum 0.01"

shape: rectangle

box: [0.534, 0.435, 0.213, 0.255]

- id: B

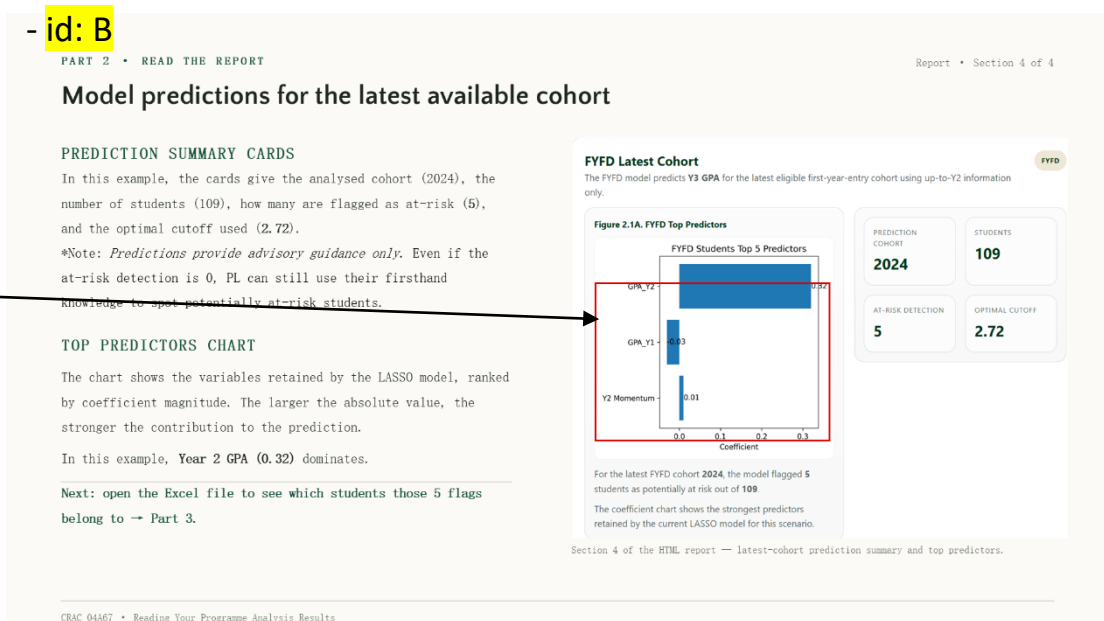
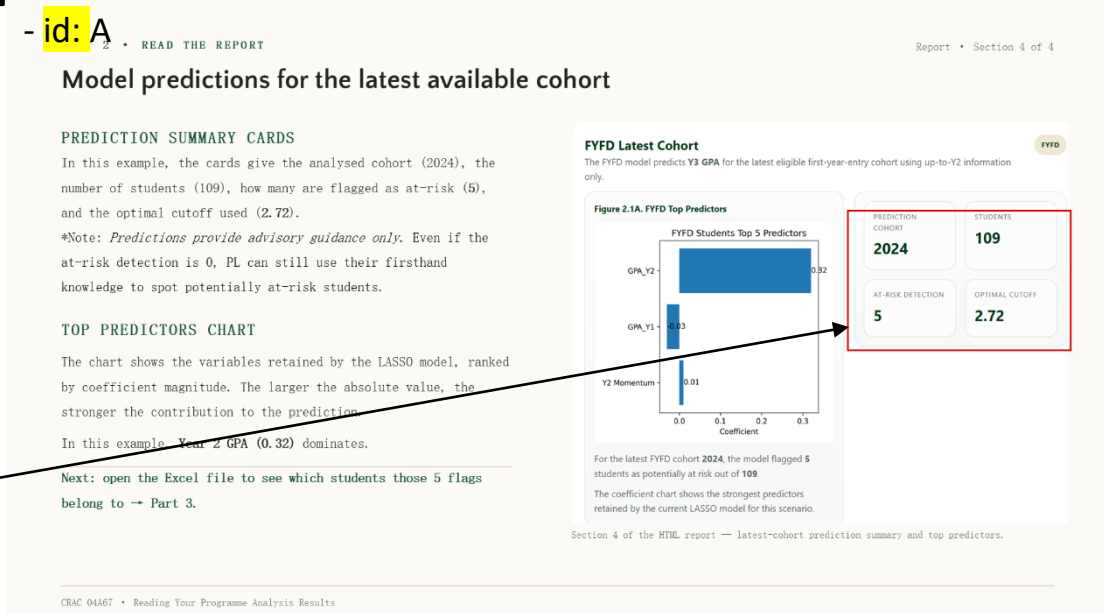
location: "Right of the chart (summary cards)"

element: "Summary cards: PREDICTION COHORT 2024 / STUDENTS 109 / AT-RISK DETECTION 5 / OPTIMAL CUTOFF 2.72"

shape: rectangle

box: [0.763, 0.343, 0.203, 0.227]

Visual Anchor Annotation(.txt,.md,.yaml)



One slide becomes several cue images

slide-12.png
base • 无标记

讲整页时使用，画面保持原样

slide-12-A.png
cue A • 框住图表

讲到模型保留的预测变量时使用

slide-12-B.png
cue B • 框住摘要卡

讲到 cohort 规模与风险人数时用

帶有视觉提示的图严格等于「原始 PNG + 按坐标画出的一个红框或红点」，没有其他改动。The visual cue image is strictly equivalent to the "original PNG plus a red box or red dot overlaid at specific coordinates," with no other modifications.

后面配音时，每一帧对应语音讲稿里的一段。 Each frame later maps to exactly one audio segment.

5. 按锚点切分讲稿，用 TTS 生成音频

Split the script by cues and synthesize speech

5. 按锚点切分讲稿, 用 TTS 生成音频 Split the script by cues and synthesize speech

Slide 12 — Report · Section 4 of 4 — Predictions for the latest cohort

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- [A] Top predictors bar chart (LASSO)
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1. 解析讲稿, 按视觉提示点切分 Parse the script
2. 选定AI音色 Choose a TTS voice (e.g., Mimo TTS)
3. 逐段合成音频 Synthesize clips
4. ffmpeg实测每段音频时长并记录 Measure duration
5. 根据每一段实测的mp3的音频时长, 设定下一步每一个png的时间参数 write the timeline

slide-12-base.mp3



The final report section moves from historical validation to predictions for the latest eligible cohort

slide-12-A.mp3



[A] The chart explains which variables the LASSO model retained. LASSO is a regression method that shrinks less useful coefficients toward zero, leaving a relatively compact set of predictors. It was chosen in part because the retained variables and their coefficients are easier to explain and audit than a more opaque model such as a neural network. The larger the absolute coefficient, the stronger the retained contribution; here, Year two GPA is clearly the dominant predictor at zero point three two.

slide-12-B.mp3



[B] In this example, the cohort is twenty twenty-four, with one hundred and nine students, and five are flagged as potentially at risk using the model's optimal cutoff of two point seven two

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"duration": 9.84},  
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"duration": 20.0 },  
{"slide": 12, "order": 3, "cue": "B", "tag": "B", "audio": "slide-12-B.mp3",  
"duration": 12.64}
```

一对一 One-to-one; 每一段 MP3 只配一张 PNG One MP3 ↔ one PNG.

Slide 12 — Report • Section 4 of 4 — Predictions for the latest cohort

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slide-12-base.mp3

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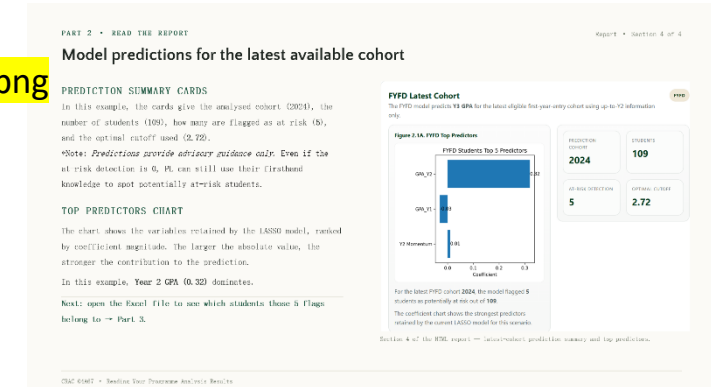
slide-12-A.mp3

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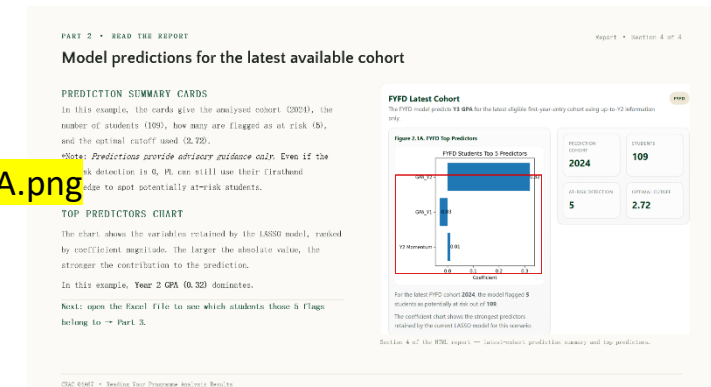
slide-12-B.mp3

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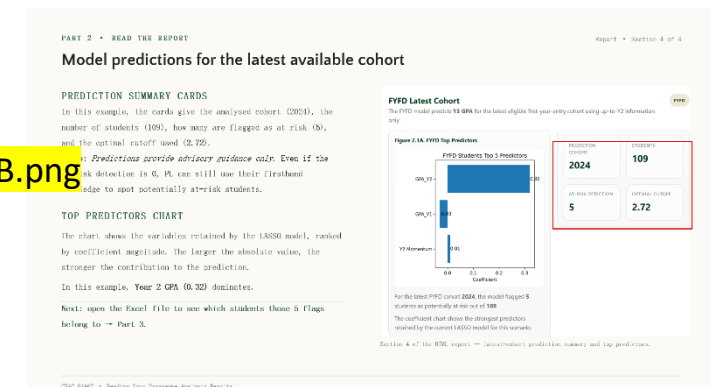
Slide-12.png



Slide-12-A.png



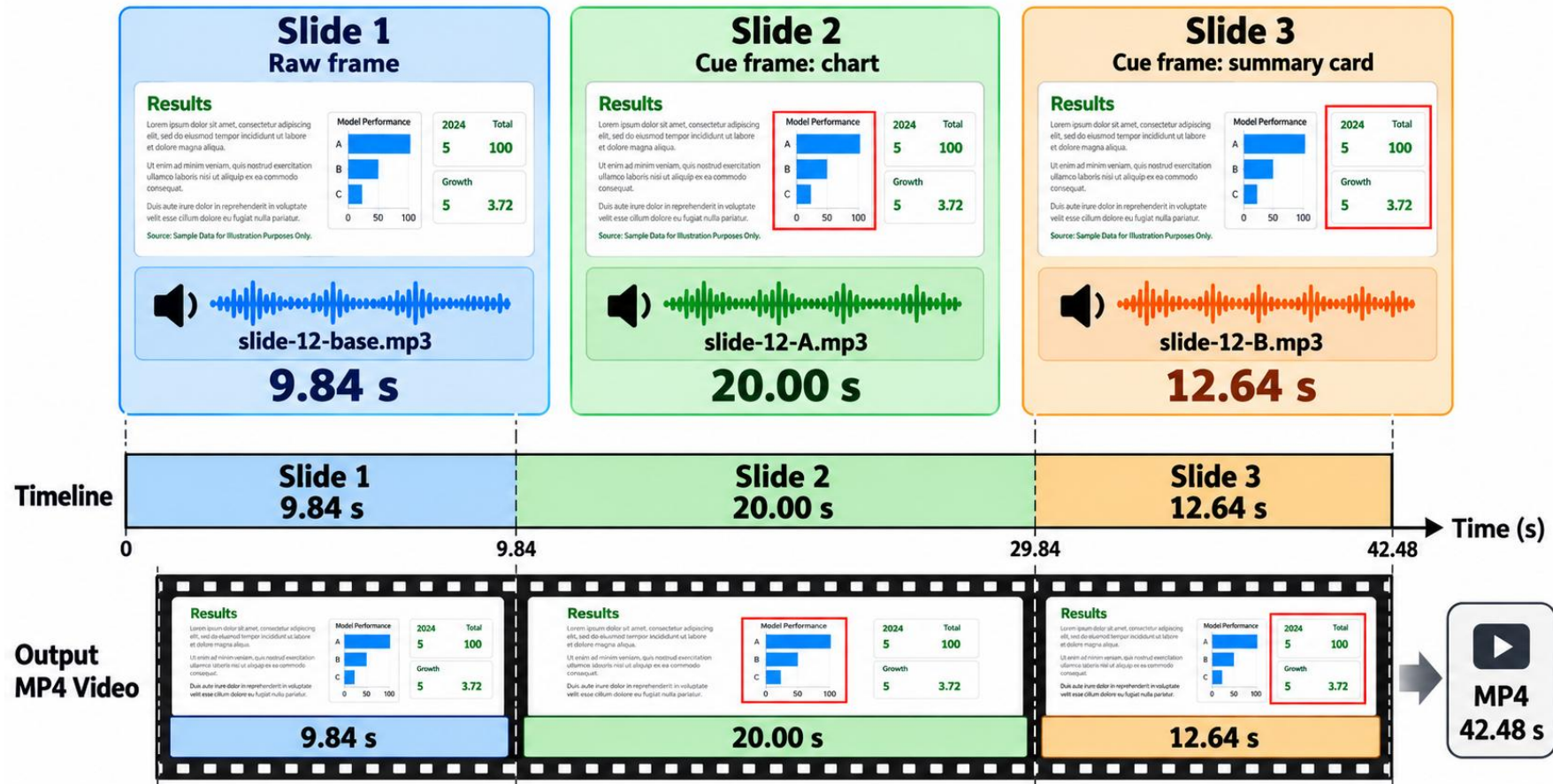
Slide-12-B.png



6. 按实测时长把图和声音合成视频 Assemble frames and audio onto one timeline

6. Assemble Video: Mux picture, voice, and duration into an MP4

视频合成：图像 + 音频 + 时长信息 → MP4



音画同步：音频时长决定对应图片的显示时间

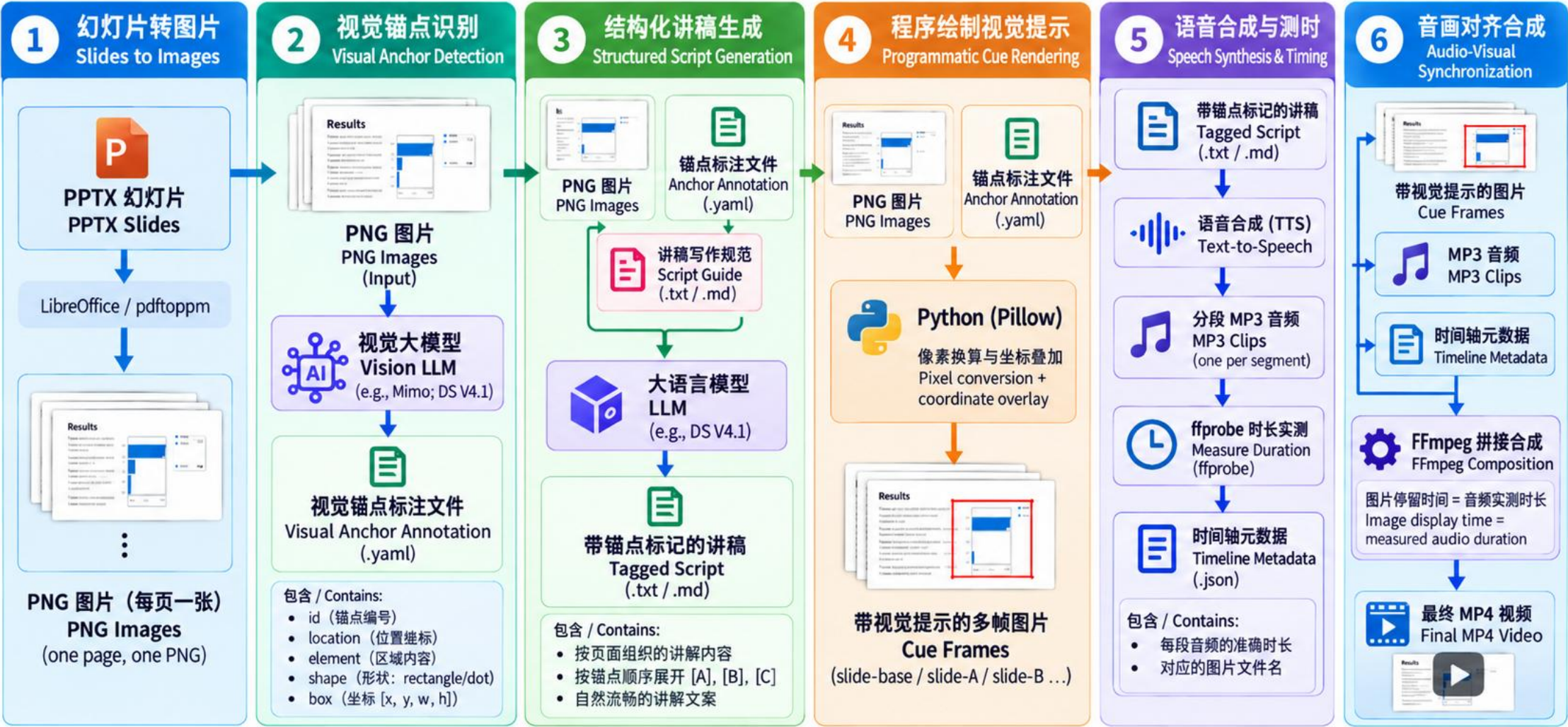
Synchronization: Audio duration determines image display time.

视频合成：使用 FFmpeg 按序合并所有素材

Concatenation: Use FFmpeg to sequentially merge media into one video.

总结 Summary

全流程架构 Methodology Summary: The 6-Step End-to-End Pipeline



AI SKILL DESIGN / AI SKILL 设计

结束END